

S12 individual evidence record

FINAL_LOCAL — content and packaging only. Native browser/PDF, exact-runtime and live service qualifications remain open. Use FORM_S12_EN_GB.html offline. S12_EVIDENCE_FORM.html and form.html are byte-identical compatibility entry points. The 49 semantic IDs and seven sections are retained: 13 core fields and 36 later fields.

Evidence and completion contract

Write the core draft during minutes 00–12. Finish required P03 implementation and actual fake-unit observations, an individually annotated P01 source/model trace and one actual bounded Gemini critique by the actual teacher-set deadline. STOP content at minute 60; the 18-minute implementation segment does not promise completion. A blocked required item remains a printable honest draft. The form grants no institutional alternative.

Student-declared evidence metadata is separate from field text. For each field named in schema.evidencePolicy.fields, record evidenceStatus[field_id] = {class, outcome}. Only exact whitelisted classes and outcomes are admitted. Required P03 metadata must describe actual PURE_JS/CANONICAL_TESTS observations and passing test results. Required P01 metadata may describe SOURCE_ANALYSIS/MODEL when individually annotated honestly. Gemini prompt, response and claim require ACTUAL_GEMINI. Independent checks have their own bounded class and outcome.

The real WebSocket integration field is a separately named qualification.

NOT_EXECUTED/BLOCKED with the actual reason does not bar the required fake-unit assessment lane. A prefixed NOT_EXECUTED in prose does not satisfy evidence metadata and cannot make a required blocked observation complete. Structural validation does not verify execution, truth, correctness or approval.

Draft migration and privacy

Use Save local draft and Export JSON. JSON imports and compact saved/exported backups are limited to 2,000,000 UTF-8 bytes, text fields to 16,000 JavaScript UTF-16 code units and plain objects to exact field/property whitelists. When indentation would exceed the import bound, export uses compact JSON while retaining all fields. An oversized aggregate draft cannot be saved/exported as a non-importable backup: the current text is retained and the failure explains manual-copy recovery. Reduce unnecessary repetition without discarding required evidence if a portable JSON backup is needed. JSON schema v1.1.0 and v1.2.0 text can be imported. Release v1.2.1 retains the compatible JSON schema v1.2.0 and local storage key TW2026_S12_FORM_1_v1.2.0; declarations, PDF review, submission, final state and all evidence metadata reset. Imported provenance remains unverified. Re-select metadata only after reviewing each actual observation.

Autosave begins OFF. Switching OFF cancels the pending save. Import, reset and erasure cancel queued saves. Reset form requires its own confirmation and clears only current text. Erase stored draft requires its own confirmation and erases only this version's application key; it retains current text, other storage and the historical v1.1.0 key. Storage failure is displayed; export JSON or copy text manually.

Remove passwords, tokens, API keys, session cookies, real personal data, private screenshots, private code and Moodle data. Use only the approved student_code/alias. The teacher keeps the private identity mapping. No surname/group fields or S11 policy are introduced.

PDF and upload route

Print honest draft is always available. Print assessed candidate requires structural assessment completeness and the declaration that the actual PDF has been reviewed. First print a truthful draft, save it, open and inspect all pages, then update the review declaration and print/review the final candidate. Never infer saving from the print dialogue. The proposed name is TW2026_S12_CODE .pdf, with CODE generated from the approved student code or alias.

Edits invalidate stale print content. Browser beforeprint rebuilds current text safely, including literal hostile text. Import and substantive edits reset declarations/PDF/submission/final state. Before upload open the saved PDF again and review filename, all text, metadata and redaction. One private S12 Assignment receives one reviewed PDF. Save changes may leave Draft; final submission requires actual observed status, filename and timestamp. See MOODLE_UPLOAD_GUIDE_S12_EN_GB.md.

Editable DOCX use: replace each blank response with your own text, expand the response area and embed your own sanitised captures with captions. This DOCX performs no structural or truth validation. Record each required evidence status explicitly; copying a status word into prose does not establish execution.

Identity and execution

Core draft, minutes 00–05. Record actual identity and execution lane before making predictions. Imported text and metadata remain unverified.

Student code or approved alias

FIELD student_code | CORE

Use only the student code or alias approved by the teacher. The teacher keeps the private identity mapping. CODE proposes TW2026_S12_CODE.pdf; do not add surname or group fields.

[Enter your own response. Blank means not yet recorded.]

Public package ID

FIELD package_id | CORE

Copy the exact 64-character ID from PACKAGE_ID.txt at the extracted student root. Run VERIFY_PACKAGE before editing; after editing use the separate one-file work boundary.

[Enter your own response. Blank means not yet recorded.]

Assessed source SHA 256

FIELD source_sha256 | CORE

Before implementation, copy source_sha256 from a successful VERIFY_INITIAL_STATE receipt and its output locator. If that gate is blocked, record NOT_RECORDED or BLOCKED with the actual reason; do not invent a hash. After required work passes, replace this entry with source_sha256 from VERIFY_WORK_RESULT and its final receipt locator before the assessed PDF. A hash identifies bytes, not correct behaviour.

[Enter your own response. Blank means not yet recorded.]

Actual Node js version

FIELD actual_node | CORE

Copy the actually displayed Node version from CHECK_ENVIRONMENT. Prescribed v24.21.0; a mismatch is a block for strict runtime qualification, not permission to relabel your version.

[Enter your own response. Blank means not yet recorded.]

Actual npm version

FIELD actual_npm | CORE

Copy the actually displayed npm version from CHECK_ENVIRONMENT. Prescribed 11.19.0. Do not install or update tools through this form.

[Enter your own response. Blank means not yet recorded.]

Evidence execution class

FIELD execution_class | CORE

Choose the strongest class actually used. PURE_JS means executable JavaScript with the supplied fake transport, MODEL means a constructed trace, REAL_PROTOCOL requires an actual protocol run. Per-field metadata below narrows individual claims.

Response options: NOT_EXECUTED / SOURCE_ANALYSIS / PURE_JS / MODEL / CANONICAL_TESTS / REAL_PROTOCOL

[Enter your own response. Blank means not yet recorded.]

Environment limitation or block

FIELD environment_limit | CORE

Record the exact available lane, block and action requested from the teacher. The real WebSocket lane is separate; no dependency installation is a hidden requirement of the fake-unit lane.

[Enter your own response. Blank means not yet recorded.]

Protocol and prediction

Core draft, minutes 05–12. Write predictions before implementation and observation; distinguish request state from lifetime subscriptions.

Acceptance execution and delivery owner map

FIELD accepted_owner_map | CORE

At 05–12 minutes predict E07: HTTP accepts, calculation executes and the registered connection receives delivery. An HTTP 202 response does not say that calculation is complete.

[Enter your own response. Blank means not yet recorded.]

Chosen transport and justification

FIELD transport_choice | CORE

Explain why a shared WebSocket channel needs correlation while the HTTP request has its own response. State whether your evidence is a fake-unit run, source analysis, model or real protocol.

[Enter your own response. Blank means not yet recorded.]

Principal connection and request identity map

FIELD connection_identity | CORE

E07: distinguish principal identity, connectionId and requestId. Locate guided/p01 source or S12_P01_GUIDED_TRACE.md. Query identity in the teaching fixture is not production authentication.

[Enter your own response. Blank means not yet recorded.]

Prediction pending before send ordering

FIELD pending_before_send_prediction | CORE

E02: predict what happens if transport.send immediately calls the message listener. State why the pending entry must already exist before send.

[Enter your own response. Blank means not yet recorded.]

Predicted terminal paths

FIELD terminal_paths_prediction | CORE

E03–E06: predict completion, remote failure, send failure, timeout, abort, close and disposal. Name the owner that removes state before settling once.

[Enter your own response. Blank means not yet recorded.]

Scoped cleanup invariant

FIELD cleanup_invariant | CORE

E04–E05: predict per-request pending/timer/abort cleanup. One dispatcher subscription pair remains until dispose; the adapter and socket have separate owners. Zero pending is a narrow claim.

[Enter your own response. Blank means not yet recorded.]

P03 implementation and evidence

Later fields, minutes 12–39 and work after STOP 60. One implementation path is assessed. Required fake-unit observations and separate real-protocol qualification have different gates.

Assessed P03 path changed

FIELD p03_changed_path | FINAL

Enter exactly projects/p03/student/src/request-dispatcher.mjs, relative to the extracted student root. This is the only assessed implementation file you edit.

[Enter your own response. Blank means not yet recorded.]

Narrow diff or hash locator

FIELD p03_diff_locator | FINAL

Locate your narrow diff and completed source hash. Explain changes relevant to E01–E06 without pasting a full answer into Gemini.

[Enter your own response. Blank means not yet recorded.]

P03 baseline results

FIELD p03_baseline_results | FINAL

Record actual baseline count, result and output locator. Select class PURE_JS or CANONICAL_TESTS and outcome PASS only after observing the completed-work verifier.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: PASS.

[Enter your own response. Blank means not yet recorded.]

P03 objective results

FIELD p03_objective_results | FINAL

Record original canonical objective checks and derived successor checks from VERIFY_WORK_RESULT. The starter direct rejection is not an intended assertion. Normal assessed completeness requires actual PASS.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: PASS.

[Enter your own response. Blank means not yet recorded.]

P03 regression results

FIELD p03_regression_results | FINAL

Record actual regression counts and output locator from VERIFY_WORK_RESULT. A parser failure, skip, cancellation, timeout or guard block is not PASS.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: PASS.

[Enter your own response. Blank means not yet recorded.]

P03 integration result or honest block

FIELD p03_integration_results | FINAL

Separate qualification: name the real ws run and limits if actually performed. Otherwise write the actual NOT_EXECUTED/BLOCKED reason and choose that metadata. This field does not require a real protocol run for the fake-unit assessment lane.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Separate real protocol qualification. Admitted classes: REAL_PROTOCOL / NOT_EXECUTED / BLOCKED. Admitted outcomes: PASS / FAIL / NOT_EXECUTED / BLOCKED / OBSERVED.

[Enter your own response. Blank means not yet recorded.]

Reversed order trace

FIELD p03_reversed_order_trace | FINAL

E01: start r1, r2 and r3; deliver r3, r1 then r2. Record which caller receives each result, the output locator and any difference from prediction.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Synchronous reply trace

FIELD p03_sync_reply_trace | FINAL

E02: record the immediate-reply fake transport case, the state-before-send ordering and the observed result. Do not describe a queued asynchronous reply as a synchronous test.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Timeout trace

FIELD p03_timeout_trace | FINAL

E04: record timeout outcome, one settlement and late completion behaviour. Name the pending/timer/abort counters and their observation point.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Abort trace

FIELD p03_abort_trace | FINAL

E03–E04: record pre-abort send count 0 and active abort cleanup. Local promise rejection does not prove cancellation of server work.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Transport close trace

FIELD p03_close_trace | FINAL

E05: record rejection of pending requests and refusal of dispatch after permanent close. Include output or test locator.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Dispose trace

FIELD p03_dispose_trace | FINAL

E05: record idempotent dispose, removal of dispatcher subscriptions and refusal of future dispatch. Adapter disposal and closing the socket are separate obligations.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Evidence that pending count returns to zero

FIELD p03_zero_pending | FINAL

E04–E05: report the actual pending count after each terminal path with a locator. Explain that zero pending alone says nothing about adapter/socket resources.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Scoped listener and timer cleanup evidence

FIELD p03_zero_listeners_timers | FINAL

E04–E05: separate per-request timer/abort listeners (zero after settlement), dispatcher subscription pair (retained until dispose) and adapter/socket ownership. Record each scope rather than invent one global zero.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: PURE_JS / CANONICAL_TESTS. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Unresolved P03 work

FIELD p03_unresolved_work | FINAL

List genuinely unresolved required obligations or explicitly state that none remain and give the completed-work result locator. The 18-minute seminar implementation segment is not guaranteed full completion.

[Enter your own response. Blank means not yet recorded.]

P01 guided trace

Later fields, minutes 50–55. Each student individually annotates the required P01 source/model trace. The labels do not turn a constructed trace into a live HTTP/WebSocket run.

P01 guided demonstration scope

FIELD p01_demo_scope | FINAL

E07 required individual guided trace: identify the supplied source/model trace or actual run. Name the source file or model locator and your individual annotations.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

HTTP acceptance observation

FIELD p01_http_acceptance | FINAL

E07: annotate the HTTP 202/requestId acceptance point and the later completion point. If source/model only, say so and select matching metadata.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

WebSocket registration observation

FIELD p01_ws_registration | FINAL

E07: annotate connection registration, acknowledgement and matching target. Identify observed versus inferred behaviour.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Targeted result observation

FIELD p01_targeted_result | FINAL

E07: trace the principal/connection/request tuple to its target. Explain why an out-of-order result must reach its own caller.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Identity binding observation

FIELD p01_identity_binding | FINAL

E07: record the binding and the limit of the query-identity teaching fixture. Do not call it production authentication.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Cleanup observation

FIELD p01_cleanup_observation | FINAL

E07: annotate pending removal at delivery and source limitations: original duplicate IDs can overwrite ownership and disconnect does not remove pending work. Source analysis is not a repaired live system.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Guided demo limitation

FIELD p01_demo_limit | FINAL

State the evidence class, source limitations and untested real HTTP/WebSocket behaviour. A source/model trace is acceptable for the required guided trace when labelled accurately.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS.

[Enter your own response. Blank means not yet recorded.]

Bounded Gemini critique

Later fields, minutes 39–50. Preserve one actual bounded critique and your independent check. Unavailable access stays a truthful draft; this form grants no alternative.

Actual bounded Gemini prompt

FIELD gemini_prompt | FINAL

Paste the one actual sanitised prompt you sent using GEMINI_PROMPT_EN_GB.txt. Include only a minimal public excerpt; no full dispatcher, credentials, private code or Moodle data.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: ACTUAL_GEMINI. Admitted outcomes: OBSERVED.

[Enter your own response. Blank means not yet recorded.]

Relevant Gemini response excerpt

FIELD gemini_response_excerpt | FINAL

Preserve the relevant actual response excerpt, model label if visible and a local locator. No invented dialogue. If access is blocked choose BLOCKED and retain a truthful draft.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: ACTUAL_GEMINI. Admitted outcomes: OBSERVED.

[Enter your own response. Blank means not yet recorded.]

Falsifiable claim evaluated

FIELD gemini_claim | FINAL

Quote or paraphrase one falsifiable response claim and distinguish Observed, Inferred or Unknown. The supplied flawed claim says pendingCount 0 proves all lifecycle cleanup.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: ACTUAL_GEMINI. Admitted outcomes: OBSERVED.

[Enter your own response. Blank means not yet recorded.]

Independent check

FIELD gemini_independent_check | FINAL

E04–E05 or a narrow source counterexample: record your own check, result, locator and limit. A Gemini answer cannot serve as its own independent check.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / CANONICAL_TESTS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS / FAIL.

[Enter your own response. Blank means not yet recorded.]

Verdict correction and limitation

FIELD gemini_verdict_correction | FINAL

State ACCEPTED, REJECTED, PARTIALLY ACCEPTED or UNKNOWN, then correction and limit. The verdict must follow your independent check, not the model confidence.

Separate evidence status metadata: class [record actual class]; outcome [record actual outcome]. This is an unverified student declaration.

Required assessment evidence. Admitted classes: SOURCE_ANALYSIS / MODEL / PURE_JS / CANONICAL_TESTS / REAL_PROTOCOL. Admitted outcomes: OBSERVED / PASS / FAIL.

[Enter your own response. Blank means not yet recorded.]

Reflection and transfer

Later fields and exit ticket, minutes 55–60. P02 is optional. Record limits and next work at STOP 60 without promising completion in the seminar.

Optional P02 status

FIELD p02_optional_status | FINAL

Choose NOT_STARTED if you did not attempt optional Redis/BullMQ work. P02 adds no assessed implementation gate and no hidden provisioning requirement.

Response options: NOT_STARTED / PLANNED / PARTIAL / COMPLETE_UNVERIFIED / COMPLETE_WITH_EVIDENCE

[Enter your own response. Blank means not yet recorded.]

Cross protocol reflection

FIELD cross_protocol_reflection | FINAL

Explain how acceptance, execution and targeted delivery differ, using your own P03 and P01 locators. Include one risk such as an accepted report export being mistaken for a completed total.

[Enter your own response. Blank means not yet recorded.]

Evidence class limitations

FIELD evidence_class_limits | FINAL

List which claims are actual fake-unit observations, source/model annotations or real protocol observations, and which remain untested.

[Enter your own response. Blank means not yet recorded.]

Required next work

FIELD next_work | FINAL

At minute 60 record the next required action, teacher-set deadline when actually known and outstanding qualification. No deadline or penalty is invented by this form.

[Enter your own response. Blank means not yet recorded.]

Declaration and PDF route

After required work and final review. Declarations renew after edits or import. Printing, PDF saving and Moodle final submission are separate observed actions.

I removed credentials tokens cookies and unnecessary personal data

FIELD privacy_redaction | FINAL

After the latest edit, inspect all fields and remove passwords, tokens, API keys, session cookies, real personal data, private account screenshots, private code and Moodle data. Then renew this declaration.

[Unchecked. Review personally before declaring.]

I declare that the evidence and limitations are truthful

FIELD declaration_truthful | FINAL

Declare only your own truthful evidence and limits. Structural completeness is not teacher approval or evidence verification. Import and substantive edits reset this declaration.

[Unchecked. Review personally before declaring.]

PDF review status

FIELD pdf_review_status | FINAL

Choose FINAL_REVIEWED only after opening the actual saved PDF and checking every page, filename, text, evidence status and redaction. Browser print alone cannot verify saving.

Response options: NOT_REVIEWED / DRAFT_REVIEWED / FINAL_REVIEWED

[Enter your own response. Blank means not yet recorded.]

Submission status

FIELD submission_status | FINAL

READY_NOT_SUBMITTED describes local readiness. SUBMITTED_UNVERIFIED records a claim you must support by observing the actual Moodle final status/name/timestamp. Draft is not final submission.

Response options: NOT_SUBMITTED / READY_NOT_SUBMITTED / SUBMITTED_UNVERIFIED

[Enter your own response. Blank means not yet recorded.]

Form status

FIELD final_status | FINAL

FINAL_FIELDS_COMPLETE_NOT_VERIFIED is a structural candidate, not correctness, teacher approval or a Moodle receipt. Keep DRAFT whenever required implementation, actual Gemini exchange or evidence remains blocked.

Response options: DRAFT / FINAL_FIELDS_COMPLETE_NOT_VERIFIED

[Enter your own response. Blank means not yet recorded.]